

Muhammad Huzaifa Khan

Assignment

Assignment # 12-13
JAVASCRIPT

1. Write a program that takes a character (number or string) in a variable & checks whether the given input is a number, uppercase letter or lower case letter. (Hint: ASCII codes:- A=65, Z=90, a=97, z=122).

Code:

```

<!DOCTYPE html>
<html>
<head>
  <title>Question 1</title>
</head>
<body>

  <h2>Character Checker</h2>

  <script>

    let ch = prompt("Enter a character:");

    let ascii = ch.charCodeAt(0);

    if (ascii >= 48 && ascii <= 57)
    {
      document.write(ch + " is a Number.");
    }
    else if (ascii >= 65 && ascii <= 90)
    {
      document.write(ch + " is an Uppercase Letter.");
    }
    else if (ascii >= 97 && ascii <= 122)
    {
      document.write(ch + " is a Lowercase Letter.");
    }
    else
    {
      document.write("Invalid Input");
    }

  </script>

</body>
</html>

```

Output:

Sample Output 1

Input:

A

Output:

A is an Uppercase Letter.

Sample Output 2

Input:

a

Output:

a is a Lowercase Letter.

Sample Output 3

Input:

7

Output:

7 is a Number.

2. Write a JavaScript program that accept two integers and display the larger. Also show if the two integers are equal.

Code:

```
<!DOCTYPE html>  
<html>  
<head>  
<title>Question 2</title>  
</head>  
<body>  
  
<h2>Larger Number</h2>  
  
<script>  
  
let num1 = Number(prompt("Enter First Number"));  
let num2 = Number(prompt("Enter Second Number"));  
  
if(num1 > num2)  
{  
document.write(num1 + " is larger.");  
}  
else if(num2 > num1)  
{  
document.write(num2 + " is larger.");  
}  
else  
{  
document.write("Both numbers are equal.");  
}  
  
</script>  
  
</body>  
</html>
```

Output:

Sample Output 1

```
Input:
12
25

Output:
25 is larger.
```

Sample Output 2

```
Input:
20
20

Output:
Both numbers are equal.
```

3. Write a program that takes input a number from user & state whether the number is positive, negative or zero.

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Question 3</title>
</head>
<body>

<h2>Positive / Negative / Zero</h2>

<script>

let number = Number(prompt("Enter any Number"));

if(number > 0)
{
document.write(number + " is Positive.");
}
```

```
else if(number < 0)  
{  
document.write(number + " is Negative.");  
}  
else  
{  
document.write("The number is Zero.");  
}  
  
</script>  
  
</body>  
</html>
```

Output:

Sample Output 1

Input:

15

Output:

15 is Positive.

Sample Output 2

Input:

-8

Output:

-8 is Negative.

Sample Output 3

Input:

0

Output:

The number is Zero.

4. Write a program that takes a character (i.e. string of length 1) and returns true if it is a vowel, false otherwise

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Question 4</title>
</head>
<body>

<h2>Vowel Checker</h2>

<script>

let ch = prompt("Enter a character:");

if(ch=="a" || ch=="e" || ch=="i" || ch=="o" || ch=="u" ||
ch=="A" || ch=="E" || ch=="I" || ch=="O" || ch=="U")
{
document.write("True<br>");
document.write(ch + " is a vowel.");
}
else
{
document.write("False<br>");
document.write(ch + " is not a vowel.");
}

</script>

</body>
</html>
```

Output:

Sample Output 1

```
Input:
a

Output:
True
a is a vowel.
```

Sample Output 2

```
Input:
k

Output:
False
k is not a vowel.
```

5. Write a program that
 - a. Store correct password in a JS variable.
 - b. Asks user to enter his/her password
 - c. Validate the two passwords:
 - i. Check if user has entered password. If not, then give message “ Please enter your password”
 - ii. Check if both passwords are same. If they are same, show message “Correct! The password you entered matches the original password”. Show

“Incorrect password” otherwise.

Code:

```
<!DOCTYPE html>  
<html>  
<head>  
<title>Question 5</title>  
</head>  
<body>  
  
<h2>Password Validation</h2>  
  
<script>  
  
let correctPassword = "admin123";  
  
let userPassword = prompt("Enter your password");  
  
if(userPassword == "")  
{  
document.write("Please enter your password.");  
}  
else if(userPassword === correctPassword)  
{  
document.write("Correct! The password you entered matches the original password.");  
}  
else  
{  
document.write("Incorrect password.");  
}  
  
</script>  
  
</body>  
</html>
```

Output:

Sample Output 1

Input:
admin123

Output:
Correct! The password you entered matches the original password.

Sample Output 2

Input:
12345

Output:
Incorrect password.

Sample Output 3

Input:
(empty)

Output:
Please enter your password.

6. This if/else statement does not work. Try to fix it:

```
var greeting;
var hour = 13;
if (hour < 18) {
  greeting = "Good day";
else
  greeting = "Good evening";
}
```

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 6</title>
</head>
<body>

  <h2>Greeting Program</h2>

  <script>

    var greeting;
    var hour = 13;

    if(hour < 18)
    {
      greeting = "Good Day";
    }
    else
    {
      greeting = "Good Evening";
    }

    document.write(greeting);

  </script>

</body>
</html>
```

Output:

Good Day

7. Write a program that takes time as input from user in 24 hours clock format like: 1900 = 7pm. Implement the following case using if, else & else if statements

Code:

```

<!DOCTYPE html>
<html>
<head>
  <title>Question 7</title>
</head>
<body>

  <h2>Greeting According to Time</h2>

  <script>

    let time = Number(prompt("Enter time in 24-hour format (e.g. 1900):"));

    if(time >= 0 && time < 1200)
    {
      document.write("Good Morning!");
    }
    else if(time >= 1200 && time < 1700)
    {
      document.write("Good Afternoon!");
    }
    else if(time >= 1700 && time < 2100)
    {
      document.write("Good Evening!");
    }
    else if(time >= 2100 && time <= 2359)
    {
      document.write("Good Night!");
    }
    else
    {
      document.write("Invalid Time");
    }

  </script>

</body>
</html>

```

Output:

Sample Output 1

Input:
900

Output:
Good Morning!

Sample Output 2

Input:
1500

Output:
Good Afternoon!

Sample Output 3

Input:
1900

Output:
Good Evening!

Sample Output 4

Input:
2200

Output:
Good Night!

